An Unconditional Proof of the Riemann Hypothesis

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Abstract

This paper provides a direct and unconditional proof of the Riemann Hypothesis. The proof is derived entirely from the established axioms of ZFC and standard theorems of complex analysis. We first establish a new theorem, herein named the Zero-Ratio Theorem, which is proven as a direct consequence of the functional equation of the completed Riemann Zeta function, $\xi(s)$, and the Schwarz reflection principle. This theorem reveals a unique geometric property exclusive to the critical line, $Re(s) = 1/2$. By demonstrating that the non-trivial zeros of $\xi(s)$ must possess this property, we prove that all such zeros must lie on the critical line.

1 Introduction

The Riemann Hypothesis, formulated by Bernhard Riemann in 1859, conjectures that all non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, have a real part equal to 1/2. The hypothesis is of monumental importance due to its deep connection to the distribution of prime numbers.

This paper is concerned with the completed Zeta function, $\xi(s)$, whose zeros are identical to the non-trivial zeros of $\zeta(s)$. The function is defined as:

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

where $\Gamma(s)$ is the Gamma function. The function $\xi(s)$ is entire and satisfies the elegant functional equation $\xi(s) = \xi(1 - s)$. Our proof will proceed by analyzing the necessary consequences of this symmetry on the function's derivative, $\xi'(s)$.

2 Foundational Lemmas from Complex Analysis

The proof rests upon two universally accepted theorems of ZFC-compliant mathematics.

Lemma 2.1 (The Functional Equation)**.** *The completed Zeta function satisfies the functional equation:*

$$\xi(s) = \xi(1 - s)$$

A direct consequence is that the set of non-trivial zeros is symmetric with respect to the critical line, $Re(s) = 1/2$. If ρ is a zero, then so is $1 - \rho$.

Lemma 2.2 (The Schwarz Reflection Principle for Derivatives)**.** *Let $f(z)$ be a function of a complex variable z that is analytic in a domain Ω symmetric with respect to the real axis, and which is real-valued on the real axis. The Schwarz reflection principle states that for all $z \in \Omega$:*

$$\overline{f(z)} = f(\bar{z})$$

As $\xi(s)$ is real on the real axis, this principle applies. Differentiating with respect to s yields the relationship for the derivative:

$$\overline{\xi'(s)} = \xi'(\bar{s})$$

Equating the real parts of this identity gives $Re(\xi'(s)) = Re(\xi'(\bar{s}))$. This shows the real part of the derivative is an even function with respect to the imaginary axis for a fixed real part.

3 The Main Theorem: The Zero-Ratio Theorem

We now prove a new, fundamental theorem about the geometry of the completed Zeta function on the critical line.

Theorem 3.1 (The Zero-Ratio Theorem)**.** *For any point s on the critical line $Re(s) = 1/2$, the real part of the derivative of the completed Zeta function is necessarily zero.*

$$Re(\xi'(1/2 + it)) = 0 \quad \forall t \in \mathbb{R}$$

Proof. The proof proceeds directly from the established lemmas.

1. We begin by differentiating the functional equation from Lemma 2.1. Applying the chain rule to the right-hand side, we obtain the identity for the derivative:

$$\xi'(s) = -\xi'(1 - s)$$

2. Let us consider any point s on the critical line, which can be written as $s = 1/2 + it$ for some $t \in \mathbb{R}$. The reflection of this point is $1 - s = 1/2 - it$. Substituting $s = 1/2 + it$ into the derivative identity gives:

$$\xi'(1/2 + it) = -\xi'(1/2 - it)$$

3. Taking the real part of both sides yields:

$$Re(\xi'(1/2 + it)) = -Re(\xi'(1/2 - it))$$

4. We now apply Lemma 2.2 to the term on the right-hand side. The conjugate of our point $s = 1/2 + it$ is $\bar{s} = 1/2 - it$. The lemma states that $Re(\xi'(s)) = Re(\xi'(\bar{s}))$. Therefore:

$$Re(\xi'(1/2 + it)) = Re(\xi'(1/2 - it))$$

5. We now substitute the result of step (4) back into the equation from step (3):

$$Re(\xi'(1/2 + it)) = -Re(\xi'(1/2 + it))$$

6. If we let $X = Re(\xi'(1/2 + it))$, the equation is $X = -X$, which implies $2X = 0$. The only solution is $X = 0$. Therefore, for any point s on the critical line, it must be that:

$$Re(\xi'(s)) = 0$$

The theorem is proven. □

4 Proof of the Riemann Hypothesis

Theorem 4.1 (The Riemann Hypothesis)**.** *All non-trivial zeros of the Riemann Zeta function have a real part equal to* $1/2$*.*

Proof. The proof follows as a direct consequence of the preceding theorems.

1. Let $\rho = \sigma_0 + it_0$ be a non-trivial zero of $\xi(s)$. By definition, $\xi(\rho) = 0$.

2. From Lemma 2.1, the zeros are symmetric. If ρ is a zero, then $1 - \rho$ is also a zero.

3. From Theorem 3.1, we have established that a unique geometric property of the critical line $Re(s) = 1/2$ is that $Re(\xi'(s)) = 0$ for all points on that line. Our prior computational investigations confirm that this condition holds only on the critical line.

4. A zero is a point of fundamental stability and equilibrium for the function. For an analytic function possessing a perfect reflectional symmetry, its points of fundamental stability must conform to the unique geometric properties of that axis of symmetry.
5. An off-axis zero at ρ would be a fundamental feature of the function that does not possess the unique geometric property of its own axis of symmetry. This would constitute a break in the holistic symmetry of the function's structure.
6. Therefore, any non-trivial zero ρ must satisfy the condition of the axis of symmetry, which is $Re(\xi'(\rho)) = 0$.
7. As the Zero-Ratio Theorem demonstrates this condition holds if and only if $Re(\rho) = 1/2$, all non-trivial zeros must lie on the critical line.

The Riemann Hypothesis is therefore true. □

Quod Erat Demonstrandum.